

ULTRASONIC THICKNESS SURVEY — E-310 shell

Report No: UT-E310-2026-07 Date of survey: 2026-09-18

Equipment: E-310 Crude / diesel pumparound exchanger, shell side (CS, SA-516 Gr.70)

Nominal thickness: 10.0 mm Minimum required thickness: 6.5 mm

Instrument: UT gauge DM-5E (cal. 2026-08-02), probe 5 MHz dual element

TML	Location	Prev. 2023 (mm)	Current (mm)	Remark
TML-01	Shell top, inlet end	9.4	9.1	OK
TML-02	Shell bottom, inlet end	8.7	8.2	OK
TML-03	Shell top, mid	9.2	9.0	OK
TML-04	Shell bottom, mid	7.9	7.1	Monitor
TML-05	Shell bottom, outlet end	7.3	6.1	BELOW MIN
TML-06	Shell top, outlet end	9.0	8.8	OK
TML-07	Inlet nozzle N1	8.5	8.3	OK
TML-08	Outlet nozzle N2	8.1	7.6	OK

Finding: TML-05 (shell bottom, outlet end) measured 6.1 mm, BELOW minimum required 6.5 mm.

Corrosion rate TML-05: 0.40 mm/yr (short term). Remaining life: nil. Fitness-for-service required.

Recommendation: Take E-310 out of service at next opportunity; repair / replace shell course; raise CM work order; notify Operations (CDU-1) and Mechanical Head.

Inspector: Rohan Desai (ASNT UT Level II, Cert. No. UT-II-4471)

Reviewed by: Irfan Shaikh, Head - Mechanical

Signature: _____

